Full Stack Assignment – 33

Assignment ID: FSD-33

Title: Project Assignment Tracking System

Duration: 3 Hours

1️⃣ Objective

Design and develop a **Project Assignment Tracking System** that allows managers to assign projects to developers and track their assignments.
The system must prevent overlapping project assignments and ensure **data persistence using a database**.

2️⃣ User Roles

MANAGER

- Register and log in to the system
- Create and manage projects
- Assign developers to projects
- View all project assignments

DEVELOPER

- Register and log in to the system
- View assigned projects
- View project timelines

3️⃣ Functional Requirements

Authentication & Authorization

- Implement user registration and login
- Use **JWT-based authentication**
- Protect all backend APIs using JWT
- Enforce **role-based access control** on APIs and frontend views

Project Management

- MANAGER can create projects with start and end dates
- Project data must be stored in the database
- Project timelines must be retrievable dynamically

Project Assignment

- MANAGER assigns developers to projects
- Assignment data must be stored in the database

- DEVELOPER can view assigned projects and timelines

4️⃣ Business Rule (Critical)

⚠️ A developer must not be assigned to overlapping projects.

If project timelines overlap with an existing assignment for the same developer, the assignment must be rejected.

5️⃣ Technical Constraints (Mandatory)

- All project and assignment data **must persist in a database**
- Hardcoded or in-memory storage is **not allowed**
- Backend APIs must reflect actual database state
- JWT authentication and role validation are mandatory
- Use environment variables for sensitive configuration

6️⃣ Frontend Requirements

- Login and registration pages
- Role-based dashboards for MANAGER and DEVELOPER
- Project creation and assignment view for MANAGER
- Assigned project listing view for DEVELOPER

UI design is not evaluated; date-conflict logic and correctness matter.

7️⃣ Submission Requirements (Strict)

GitHub Repository

- Backend source code
- Frontend source code
- Proper folder structure
- Clear and meaningful commit history

Live Deployment

- Live frontend deployment link
- Live backend deployment link
- Backend must be connected to the database

README.md (Mandatory)

The README file must include:

- Project overview
- Tech stack used

- User roles and permissions
- API endpoints
- Database schema or collections
- Steps to run the project locally
- Live frontend and backend links

8️⃣ Evaluation Note

Evaluation focuses on **date-overlap validation, role enforcement, and end-to-end execution**.